

Seed bankroll proposal: \$20,000 to grow across seasons

Prepared September 17, 2026, revised the same day after outside review and a change of frame. This is a private request to family for a **gift** that seeds a betting bankroll. Nothing is owed back. It is not an investment, not a loan, and not a security. The bankroll is meant to **grow across seasons**: the rest of 2026, all of 2027, and golf strategies once they have a graded record. What follows is what the money funds, what has been earned so far, how bets are sized, and what the model says is likely to happen.

Every number below comes out of a script in this repository. Reproduce with:

```
python research/bankroll/scripts/bankroll_config_sweep.py --paths 50000 --out research/bankroll/docs
python research/bankroll/scripts/bankroll_config_sweep.py --paths 50000 --seasons 2
python research/bankroll/scripts/pooled_growth_chart.py --paths 100000 --seasons 2 --flat-stakes --out research/bankroll/docs
python research/bankroll/scripts/bankroll_stress.py --paths 50000 --out research/bankroll/docs
```

1. The ask

item	value
amount	\$20,000
horizon	the rest of 2026 (weeks 4–15, September 24 to December 12), then 2027 and onward. Bowls excluded. Week 3 (this weekend) is not in the projection
what it funds	two totals strategies already running; units re-sized off the bankroll each Monday; golf added once graded
expected bets, rest of 2026	~118: 6–12 Greenline unders a week (~107) and ~11 over-zero overs
expected bets, 2027	~250: ~134 Greenline unders and ~40 over-zero overs
recommended unit today	Greenline 1% of bankroll per bet (\$200 at the start), over-zero 1% (\$200). Re-derived weekly by the rule in §6
what happens to profit	stays in the bankroll

2. The answer in one paragraph

Median outcome is +8% by the end of 2026 and +21% by the end of 2027. About three seasons in ten end down; one in four two-season runs ends down. Fewer than 1% of runs lose a quarter of the money by the end of 2026, and none goes to zero. The unit is the largest that keeps that quarter-loss chance under 3% per season, and it is three-quarters of what fractional Kelly would allow. The main strategy's win rate is the planning number in a bracket; the bracket is reported everywhere.

horizon	unit	median	90% band	P(down)	P(–25%)	busts
rest of 2026	1%	\$21,638 (+8.2%)	\$16,919 – \$27,352	30.0%	344 of 50,000 (0.7%)	0 of 50,000
rest of 2026	0.5%	\$20,908 (+4.5%)	\$18,404 – \$23,641	28.4%	1 of 50,000	0 of 50,000
through 2027	1%	\$24,207 (+21.0%)	\$15,618 – \$36,819	23.4%	3.6%	0 of 50,000
through 2027	0.5%	\$22,435 (+12.2%)	\$17,673 – \$28,282	21.4%	0.3%	0 of 50,000

Planning prior throughout; the bracket is in §4 and §7.

3. What gets bet

Two strategies. One bankroll. Units re-sized off the bankroll at the start of each week, flat within the week.

Over-zero (floor-bias OVERs)

A model built in this repository. It finds games where the market total is pinned too low by the way books price a heavy favorite's opponent, and bets the OVER when the expected bias clears a threshold.

fact	value	source
walk-forward record, 2016–2025	151–83, 64.5% (95% CI 58.2–70.4%)	models/over_zero/docs/ROI_HITRATE.md
ROI at –110	+23.2% (CI +11.1 to +34.4%)	same
planning win rate	58.2% , the interval's lower endpoint, because the threshold was picked on this data	MODEL_GUIDE.md
price rule	–120 or better	same
bets left in 2026	~ 11 . Volume is front-loaded into weeks 1–3, already played	mc-combined-totals-2026-09-17.md
bets in a full season	30–51 (2022–25), ~68% in weeks 1–3	ROI_HITRATE.md
2026 so far	17–8 through week 2	site/lib/weekly-results.json

The better-evidenced edge. Nearly spent for 2026; it is roughly a third of 2027's expected profit.

Greenline totals (PFF vendor flags, unders)

PFF publishes a projection per game and flags totals where it disagrees with the market. Flags are captured Wednesday, graded Monday. ~85% of flags are unders.

fact	value	source
2026 graded flags	27–22, 55.1% (CI 41–68%), one week	greenline-season-review-2026-09-16.md
2023–25 personal unders, mostly the same flags	114–87, 56.7% (CI 49.8–63.4%), 201 bets	bet-history-analysis-2023-2025.md
planning prior	2026 flags plus the 2023–25 unders at half weight : 84.5–65.5, mean 56.1% , P(losing) 15%	mc_combined_totals.py
planned volume	6–12 unders a week , ~107 over the rest of 2026, ~134 in 2027	bankroll_config_sweep.py
price	–110, break-even 52.38%	same

Conflict rule

Week 2, Rice @ Notre Dame: over-zero said OVER 54.5 and Greenline flagged UNDER 55.5. Betting both pays juice twice for a hedge. **Rule: over-zero takes the game, Greenline skips it.** The skipped Greenline flag is marked not bet in the ledger with a note, so the control still grades it.

Excluded, for now

The pred-tracker-model in `research/spread/` is research, not a bet. Greenline spreads and moneylines went 21–28 and 21–25 in week 2 and are not funded. **Golf** enters at zero until it has a graded record, then at the same rule as every other leg.

4. The planning prior and its bracket

The 2023–25 unders were mostly Greenline flags in earlier seasons, bet by the same person. That is prior evidence for the same signal, not independent confirmation. They also sat six points higher in total than the 2026 flags (median 58.5 vs 52.5), so counting them in full probably overstates and ignoring them throws away 201 bets.

The planning prior counts them at half weight ($\kappa = 0.5$). Every recommendation in this document is made on it. The two endpoints are the bracket and are reported alongside so the reader can see what the choice is worth.

prior	record	mean	P(true rate below break-even)	median at 1%, rest of 2026	P(-25%) at 1%
n49, 2026 flags only	27–22	55.0%	35%	\$21,157 (+5.8%)	3.6%
planning, $\kappa = 0.5$	84.5–65.5	56.1%	15%	\$21,638 (+8.2%)	0.7%
pooled, $\kappa = 1$	141–109	56.4%	10%	\$21,714 (+8.6%)	0.4%

5. How the projection works

One simulated path is one run through every remaining week. 50,000 paths per cell of the sweep, 100,000 for the fan charts.

- **Win rates are never fixed.** Each path draws its own true win rate from the Beta posterior of the record, so the spread of outcomes includes not knowing the true rate, not just luck.
- **Bets on the same Saturday are correlated** through one scoring-environment shock (Gaussian copula, $\rho = 0.10$ assumed; 0 to 0.5 in the stress tests). Over-zero is all overs and Greenline mostly unders, so a high-scoring day helps one and hurts the other.
- **Volume is the plan: 6–12 unders a week**, drawn uniformly and capped by the week's FBS-vs-FBS slate (56–67 games a week, 9 in championship week). Over-zero resamples its history: ~11 bets for the rest of 2026, 30–51 in a full season with two-thirds in weeks 1–3.
- **Units re-sized off the bankroll each Monday**, flat within the week because Saturday kickoffs are simultaneous. No stop-loss; a path at zero stops.
- Pushes not modeled: zero realized on all 484 graded bets, all on half-point lines.
- **2027 is modeled as 2025's schedule** with the same win-rate draws. It assumes the edge persists and nothing about it is re-estimated. That is the assumption a multi-season projection cannot avoid; the weekly rule in §6 is how it gets corrected.

Full method, written for outside review: [mc-method-2026-09-17.md](#).

6. Choosing the unit

For a bankroll meant to grow, the right objective is expected log growth and the right frame is fractional Kelly. The constraint is a per-season drawdown cap.

The rule. Each Monday, the Greenline unit is the smaller of:

1. **quarter Kelly off the planning prior**, shrunk for 9 simultaneous bets that share an outcome correlation of ~0.06: today **1.3%** (single-bet quarter Kelly is 2.0%);
2. **the largest unit at which no more than 3% of seasons end down 25% or more**, with no busts, under the planning prior: today **1.0%**.

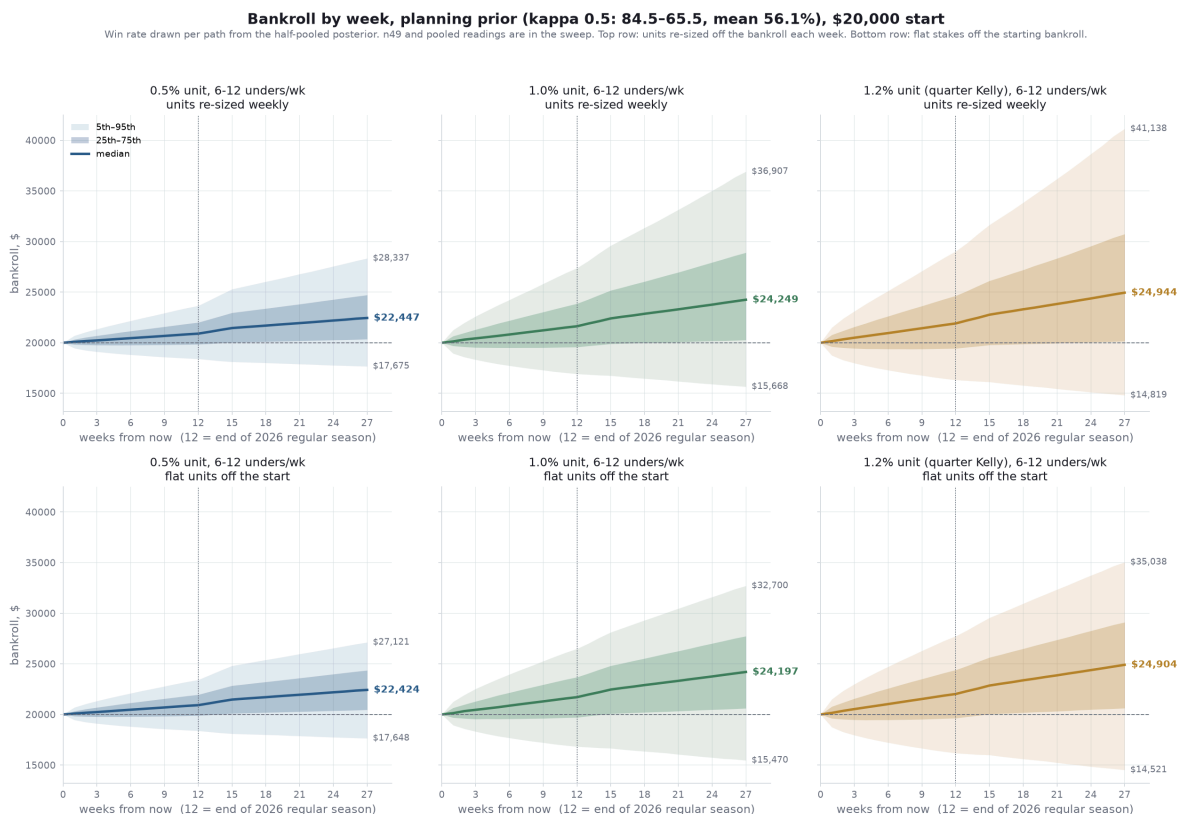
Over-zero stays at 1%. The cap binds first, so the unit is **1%**. It will rise toward 1.3% as the 2026 flags tighten the prior, and fall if they sour.

Unit sweep, rest of 2026, planning prior (bracket columns in the sweep doc):

unit	median	5th pct	95th pct	P(down)	P(-25%)	max drawdown, 95th pct	passes 3% cap
0.25%	\$20,530 (+2.6%)	\$19,032	\$22,088	28.5%	0.0%	—	yes

unit	median	5th pct	95th pct	P(down)	P(−25%)	max drawdown, 95th pct	passes 3% cap
0.50%	\$20,908 (+4.5%)	\$18,404	\$23,641	28.4%	0.0%	\$2,110	yes
1.00%	\$21,638 (+8.2%)	\$16,919	\$27,352	30.0%	0.7%	\$4,024	yes
1.2% (≈ quarter Kelly)	\$21,911 (+9.6%)	\$16,322	—	30.6%	1.8%	\$4,797	yes, but fails the combined stress case
1.50%	\$22,309 (+11.5%)	\$15,430	\$31,640	31.3%	3.8%	—	no

Through 2027 the same units give medians of +12% (0.5%), +21% (1%), +25% (1.2%); the 2-season P(−25%) at 1% is 3.6%, which is why the rule re-derives each week rather than fixing 1% for two seasons.



Three things the sweep says:

- Unit size does not change the downside ratio.** Median gain ÷ (median – 5th pct) runs 0.34–0.36 at every unit under the planning prior. Staking more buys a bigger median and a bigger 5th-percentile loss in the same proportion.
- Volume is what improves the ratio** (0.36 for one season, 0.51 for two), because more bets average out the draw of the win rate. 9 a week is ~16% of flags, above the 13% the record was earned at, so a slice of every week's bets is on flags the record never covered. The bet ledger is what prices that.
- Weekly re-sizing is nearly neutral within a season** (median −\$20 to −\$170 against flat units) and is what lets the bankroll compound across seasons.

7. Risk, stated plainly

At the recommended 1% / 1% units, rest of 2026:

measure	planning prior	n49 bracket	pooled bracket
P(season ends below \$20,000)	30.0%	38.5%	27.3%

measure	planning prior	n49 bracket	pooled bracket
P(ends below \$15,000)	0.7%	3.6%	0.4%
P(passes through \$0)	0 of 50,000	0 of 50,000	0 of 50,000
5th-percentile ending bankroll	\$16,919	\$15,432	\$17,296
mean of the worst 5% of endings	\$15,913	\$14,283	\$16,310
largest mid-season drawdown, median / 95th pct	\$1,595 / \$4,024	\$1,734 / \$5,034	\$1,571 / \$3,795
P(a drawdown deeper than 10% at some point)	35.6%	42.5%	33.9%
P(a drawdown deeper than 20%)	5.1%	11.4%	3.9%
weeks below the start, median	2	3	2
worst single week, median	−\$1,053	−\$1,075	−\$1,047
total staked over 12 weeks	~\$24,500	~\$24,300	~\$24,600

At 1% a season will spend time under water and a 10% drawdown happens in about a third of seasons. That is the price of the growth frame; 0.5% halves both.

The recommendation was stress-tested. An outside review asked whether the unit holds when the over-zero haircut is uncertain, the Greenline prior is weaker, bets 7–12 each week are worse, and same-slate correlation is up to five times the assumed value. Under the 3% cap: 0.5% passes all 25 scenarios; **1% passes 15 of 25**, failing when the prior is n49 alone or $p \geq 0.35$; 1.2% passes 13. 1% passes the combined skeptical case built on the planning prior ($P(-25\%)$ 1.7%). Details in [bankroll-stress-2026-09-17.md](#).

8. What this does not support

- **Greenline as independently validated.** $n=49$ in 2026, CI 41–68%. The planning prior is half personal history of the same signal.
- **The 2027 numbers as a forecast.** They assume the edge persists unchanged on 2025's schedule. They are what the current record implies, not what will happen.
- **Golf.** No record, no price, no volume. It is a placeholder leg.
- **A reproducible selection rule.** "Bet 6–12 of the week's flags" is a volume plan. No script picks which ones. The historical picks were hand-filtered and price-shopped.
- **Kelly as the operative rule today.** Quarter Kelly is 1.3%; the drawdown cap binds at 1%. The Kelly number is where the unit is headed if the record holds.
- **The negative cross-leg correlation as a hedge.** The shared shock is scoring environment. A common model-or-market failure that hurts both legs is not modeled.

9. What the money does each week, and the updating rule

day	step	command
Wednesday	capture Greenline flags; seed the bet ledger	<code>pull_pff_scoreboard.py --greenline;</code> <code>greenline_bet_log.py --seed</code>
Thursday–Saturday	bet 6–12 unders at −110 or better, over-zero board at −120 or better	<code>models/over_zero site</code>
Monday	grade flags; mark which were bet	<code>grade_greenline.py;</code> <code>greenline_bet_log.py</code> <code>--mark</code>
Monday	re-derive the unit: min(quarter Kelly off the planning posterior, largest unit under the 3% cap); re-size off the bankroll	<code>bankroll_config_sweep.py</code>

Updating rule, fixed now. The planning prior is the 2026 record plus the 2023–25 unders at half weight, always. The unit is re-derived each Monday by the rule in §6 and applied to the following week. No other in-season changes. A new strategy (golf) enters only with a graded record and at the same rule. Marking which flags get bet is the one manual step, and it is what turns "6–12 a week" from a plan into evidence.

Data and dates

over-zero: 234 walk-forward bets 2016–2025, `models/over_zero/docs/backtest_bets.csv`; full-season counts 2022–25 from `ROI_HITRATE.md`. Greenline: 49 graded flags, 2026 week 2, `data/ingest/pff_scoreboard/greenline_graded.csv`; 201 personal unders 2023-08 to 2025-12, `data/ingest/bet_history/history.csv`. Slate per week from `core.fact_game` FBS-vs-FBS counts: 2026 weeks 4–13, 2025 for weeks 14–15 and for the 2027 season shape. Seeds 20260917. Sweep: 50,000 paths per cell, 15 cells per horizon.

Related: [mc-method-2026-09-17.md](#), [bankroll-stress-2026-09-17.md](#), [bankroll-config-sweep-2026-09-17.md](#), [pooled-bankroll-growth-2026-09-17.md](#), [mc-combined-totals-2026-09-17.md](#).